

Jeffrey A. Oregero - Curriculum vitae

Department of Mathematics

University of Kansas

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January 16, 2026

Education

University at Buffalo, State University of New York

Department of Mathematics

2021: Ph.D. in Mathematics

Dissertation: *The focusing nonlinear Schrödinger equation with periodic boundary conditions: Spectral theory and semiclassical dynamics*

Advisor: Gino Biondini

2015: M.A. in Mathematics

Department of Economics

2013: M.A. in Economics

Ramapo College of New Jersey

Anisfield School of Business

2010: B.S. in Finance (*summa cum laude*)

Employment

University of Kansas

Department of Mathematics

Lawrence, KS 66045, USA

Adams Visiting Assistant Professor

Aug. 2023–

University of Central Florida

Department of Mathematics

Orlando, FL 32816, USA

Postdoctoral Scholar

Jan. 2022–July 2023

Simons Laufer Mathematical Sciences Institute (formerly MSRI)

Berkeley, CA 94720, USA

Postdoctoral Fellowship

Aug. 2021–Dec. 2021

Visiting appointments

Isaac Newton Institute (INI)

Cambridge, UK

Visiting Scholar

Aug. 2022–Sept. 2022

University of California, Berkeley
Berkeley, CA 94720, USA

Visiting Scholar

Aug. 2021–Dec. 2021

Research interests

Analysis of nonlinear dispersive equations with an emphasis on integrability, spectral and inverse spectral theory, singular asymptotics, stability, and algebro-geometric solutions

Publications

1. “On the modulation of wave trains in the Ostrovsky equation”, M. A. Johnson, J. Oregero, W. P. Perkins, (submitted), [arXiv:2505.21466](#)
2. “Spectral enclosure estimates for non-self-adjoint Dirac operators”, J. Oregero, (submitted), [arXiv:2504.02236](#)
3. “On the proximal dynamics between integrable and non-integrable members of a generalized Korteweg-de Vries family of equations”, N. I. Karachalios, D. Mantzavinos, and J. Oregero, (to appear–Physica D: Nonlinear Phenomena), [arXiv:2511.07609](#)
4. “Experimental observation of the spatio-temporal dynamics of breather gases in a recirculating fiber loop”, F. Copie, G. Biondini, J. Oregero, G. A. El, P. Suret, S. Randoux, *Optics Letters* **50** (22): 7043–746 (2025), [DOI 10.1364/OL.572781](#) (**Editor’s pick**)
5. “Modulational stability of wave trains in the Camassa-Holm equation”, M. A. Johnson and J. Oregero, *J. Diff. Eqs.* **446**, 113627 (2025), [DOI 10.1016/j.jde.2025.113627](#)
6. “Breather gas fission from elliptic potentials in self-focusing media”, G. Biondini, G. A. El, X.-D. Luo, J. Oregero, A. Tovbis, *Phys. Rev. E* **111**, 014214 (2025), [DOI 10.1103/PhysRevE.111.014204](#)
7. “Elliptic finite-band potentials of a non-self-adjoint Dirac operator”, G. Biondini, X.-D. Luo, J. Oregero and A. Tovbis, *Adv. in Math.* **429**, 109188 (2023), [DOI 10.1016/j.aim.2023.109188](#)
8. “On the spectrum of the periodic focusing Zakharov-Shabat operator”, G. Biondini, J. Oregero, A. Tovbis, *J. Spectr. Theory* **12** (3): 939–992 (2022), [DOI 10.4171/JST/432](#)
9. “Semiclassical dynamics and coherent soliton condensates in self-focusing nonlinear media with periodic initial conditions”, G. Biondini and J. Oregero, *Stud. Appl. Math.* **145** (3): 325–356 (2020), [DOI 10.1111/sapm.12321](#)

Seminars

1. *Soliton gases and nonlinear dispersive equations*,
Differential equations, Dynamical systems and Geometric analysis Seminar, University of Kansas, KS, Oct. 11, 2023
2. *The focusing nonlinear Schrödinger equation on the circle: an analytic approach*,
Differential equations, Dynamical systems and Geometric analysis Seminar, University of Kansas, KS, March 29, 2023

3. *Spectral theory of non-self-adjoint Dirac operators on the circle*,
Universality and Integrability in Random Matrix Theory and Interacting Particle Systems Seminar,
Simons Laufer Mathematical Sciences Institute, Berkeley, CA, Oct. 15, 2021
4. *The focusing nonlinear Schrödinger equation on the circle: spectral theory, elliptic finite-gap potentials, and soliton gases*,
Integrable Systems and Random Matrix Theory Seminar, University of Michigan, MI, Oct. 11, 2021

Presentations

Invited talks:

1. *Soliton and breather gases in self-focusing media*,
SIAM TXLA 2025 Sectional Meeting, University of Texas at Austin, TX, September 26–28, 2025
2. *The Whitham modulation system for the Camassa-Holm equation: a functional analytic approach*,
AMS Central Sectional Meeting, University of Kansas, KS, March 29–30, 2025
3. *Spectral validation of the Whitham modulation equations for the Camassa-Holm equation*,
AMS Joint Mathematics Meeting (JMM), Seattle, WA, Jan. 8–11, 2025
4. *Recent developments in the spectral theory of soliton gases: soliton and breather gases of focusing systems of AKNS type*,
SIAM Conference on Nonlinear Waves and Coherent Structures, Baltimore, MD, June 23–27, 2024
5. *Periodic gases in nonlinear dispersive hydrodynamics*,
AMS Eastern Sectional Meeting, University at Buffalo, NY, Sept. 9–10, 2023
6. *Soliton gases, breather gases, and finite-gap solutions of integrable nonlinear wave equations*,
SIAM Conference on Nonlinear Waves and Coherent Structures, University of Bremen, Germany, Aug. 30–Sept. 2, 2022
7. *The focusing Zakharov-Shabat eigenvalue problem and elliptic finite-band potentials*,
12th IMACS Conference on Nonlinear Evolution Equations and Wave Phenomena, University of Georgia, GA, March 30–April 1, 2022
8. *Semiclassical Lax spectrum of Zakharov-Shabat systems with periodic potentials*,
11th IMACS Conference on Nonlinear Evolution Equations and Wave Phenomena, University of Georgia, GA, April 17–19, 2019
9. *Small-dispersion limits for focusing NLS with periodic boundary conditions*,
SIAM Conference on Nonlinear Waves and Coherent Structures, Orange, CA, June 11–14, 2018

Contributed talks:

10. *Periodic gases in nonlinear dispersive hydrodynamics*,
19th Prairie Analysis Seminar, Kansas State University, KS, Nov. 3–4, 2023
11. *Spectral theory of a non-self-adjoint Dirac operator with a Jacobi elliptic potential*,
18th Prairie Analysis Seminar, University of Kansas, KS, Oct. 28–29, 2022
12. *Zakharov-Shabat systems with periodic potentials*,
Applied Math Days, Rensselaer Polytechnic Institute, NY, April 5–6, 2019
13. *Small dispersion limits of the focusing nonlinear Schrödinger equation with periodic boundary conditions*,
Applied Math Days, Rensselaer Polytechnic Institute, NY, April 6–7, 2018

Posters:

14. *Semiclassical Floquet spectrum of periodic Zakharov-Shabat systems*,
Workshop on Dispersive PDEs and Inverse Scattering, Fields Institute, Toronto, May 21–24, 2019

Awards and honors

- Postdoctoral Fellowship, Simons Laufer Mathematical Sciences Institute, \$33,000 (2021)
- Doctoral Dissertation Fellowship, University at Buffalo, SUNY, \$10,000 (2020)
- Student travel grant, Society for Industrial and Applied Mathematics, \$650 (2018)
- Student travel grant, The Fields Institute, \$1000 (2017)
- Ford Foundation Scholarship, \$10,000 (2010)
- Member of *Delta Mu Delta* International Business Administration Honor Society
- Member of *Golden Key* International Honour Society for Academic Excellence

Teaching

Department of Mathematics, University of Kansas

- MATH 950 Partial Differential Equations Fall 2025
- MATH 220 Applied Ordinary Differential Equations Spring 2025
- MATH 220 Applied Ordinary Differential Equations
- MATH 699 Directed Reading Fall 2024
- MATH 647 Applied Partial Differential Equations
- MATH 220 Applied Ordinary Differential Equations Spring 2024
- MATH 647 Applied Partial Differential Equations
- MATH 320 Elementary Differential Equations Fall 2023

Department of Mathematics, University of Central Florida

- MAS 3105 Matrix and Linear Algebra Spring 2023
- MAP 4303 Ordinary Differential Equations II Fall 2022
- MAP 2302 Ordinary Differential Equations I Spring 2022
- MAP 2302 Ordinary Differential Equations I Spring 2022

Simons Laufer Mathematical Sciences Institute (formerly MSRI)

- No teaching (postdoctoral fellowship) Fall 2021

Department of Mathematics, University at Buffalo, SUNY

- Mathematical Finance Instructor
- Introduction to Differential Equations
- College calculus I, II
- Survey of Calculus and Its Applications I, II
- Survey of Partial Differential Equations Teaching Assistant
- Introduction to Differential Equations
- Introduction to Linear Algebra
- College Calculus I, III
- Calculus for Business Students

Student mentoring

Department of Mathematics, University of Kansas

1. Caleb Jones
“Isoperimetric and Sturm–Liouville equations”
M.A. Thesis, committee member, successfully defended on December 5, 2025
2. Bennett Kinder
“Initial-boundary value problems and the unified transform”
Undergraduate project, project advisor, fall 2024–spring 2025
3. Haley Cabrera
“Fitting empirical dynamical models to describe the effect of climate change on population dynamics of fish species in the North Atlantic”
Undergraduate Research Award in Mathematics, project advisor, fall 2023

Professional service

Journal referee:

- SIAM Journal of Mathematical Analysis
- Advances in Mathematics
- Journal of Mathematical Physics
- Studies in Applied Mathematics
- Proceedings of the Royal Society A
- Journal of Nonlinear Science
- Physica D
- Journal of Nonlinear Waves
- Journal of Applied Mathematics and Physics (ZAMP)
- Applied Math Letters
- European Physical Journal Plus

Workshop/Seminar/Minisymposium Organization:

- Seminar: Differential equations, dynamical systems, and geometric analysis seminar, Department of Mathematics, University of Kansas, KS, Spring 2025
- Minisymposium: Recent advances in the analysis of integrable systems, AMS Central Sectional Meeting, University of Kansas, KS, March 29–30, 2025
- Minisymposium: Recent developments in dispersive partial differential equations, SIAM Nonlinear Waves and Coherent Structures, Baltimore, MD, June 23–27, 2024
- UCF/USF Workshop: Complex analytic methods with applications in orthogonal polynomials, integrable systems, and random matrix theory, University of Central Florida, Orlando, FL, Feb. 25–26, 2023

Central Florida Math Circle (<https://sciences.ucf.edu/math/circle/>)

- Lead the advanced group lessons (six weeks)

Spring 2023

- Public lesson: *Patterns in mathematics and nature* Feb. 4, 2023
- Public lesson: *Patterns in mathematics and nature* March 5, 2022

Simons Laufer Mathematical Sciences Institute Postdoctoral Fellowship Program

- Organized a series of professional development seminars for postdoctoral fellows Fall 2021

State University of New York at Buffalo Graduate Student Association (GSA)

- Treasurer–Department of Mathematics Sept. 2020–Aug. 2021

Professional skills

- Python
- Mathematica
- Matlab
- $\text{\LaTeX}$
- R

Workshops and professional development

Isaac Newton Institute of Mathematical Sciences (INI) programme: Dispersive hydrodynamics: mathematics, simulation and experiments, with applications in nonlinear waves
(visit Aug. 2022–Sept. 2022)

- (i) Analysis of dispersive systems,
Sept. 5–9, 2022
- (ii) Integrable systems and applications,
Sept. 12–16, 2022

National Science Foundation (NSF) MPS Workshop for New and Future PIs
Westin Alexandria Old Town, VA, on June 13–14 2022

Simons Laufer Mathematical Sciences Institute thematic program: Universality and Integrability in Random Matrix Theory and Interacting Particle Systems
(visit Aug. 2021–Dec. 2021)

- (i) Universality and Integrability in Random Matrix Theory and Interacting Particle Systems, Part 1,
Aug. 23–27, 2021
- (ii) Universality and Integrability in Random Matrix Theory and Interacting Particle Systems, Part 2,
Sept. 20–24, 2021
- (iii) Integrable Structures in Random Matrix Theory and Beyond,
Oct. 18–22, 2021

Fields Institute focus program: Nonlinear Dispersive Partial Differential Equations and Inverse Scattering
(visit July 31–Aug. 12 2017)

- (i) Summer School on Nonlinear Dispersive PDEs and Inverse Scattering,
July 31–Aug. 4, 2017
- (ii) Workshop on Inverse Scattering and Dispersive PDEs in One Space Dimension,
Aug. 8–11, 2017